



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating column

Key features: demographics, purchase details, shopping behavior, subscription status.

Data Preparation in Python



→ Cleaned & Standardized

Snake_case columns, median imputation for Review Rating

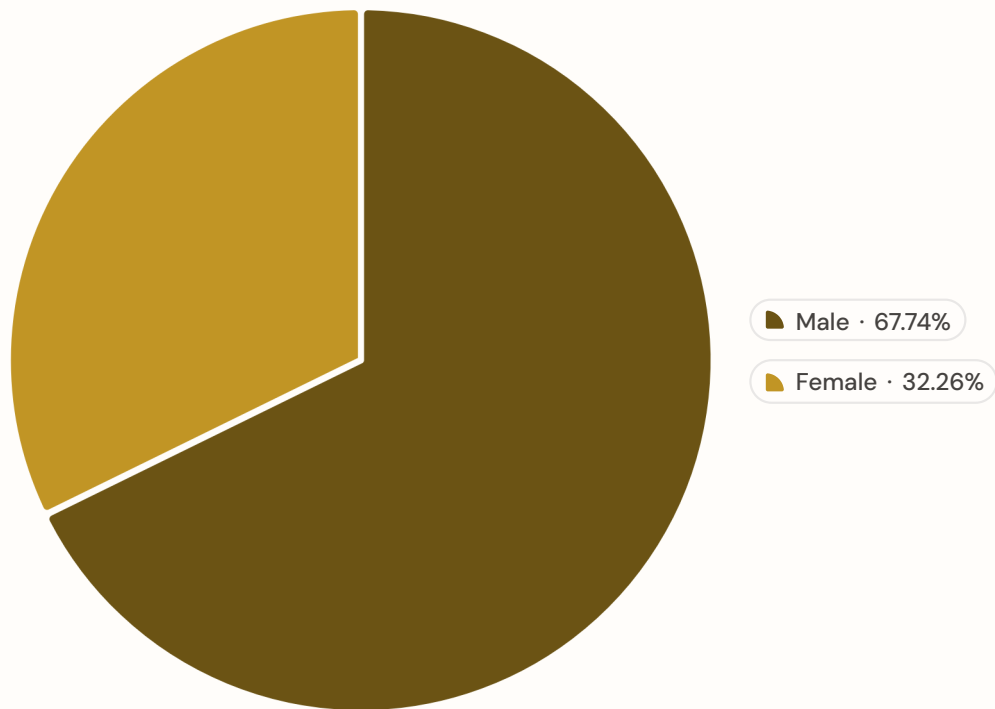
→ Feature Engineering

Created **age_group** and **purchase_frequency_days**

→ Database Integration

Loaded cleaned data into **PostgreSQL** for SQL analysis

Revenue by Gender



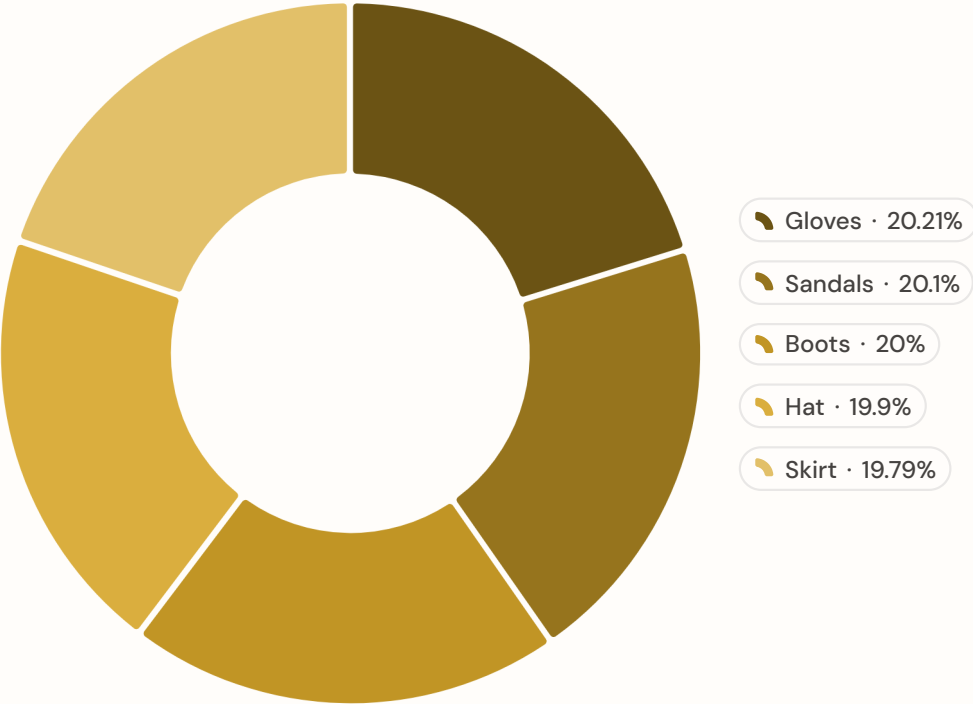
Key Insight

Male customers generated **2.1× more revenue** than female customers across all categories.

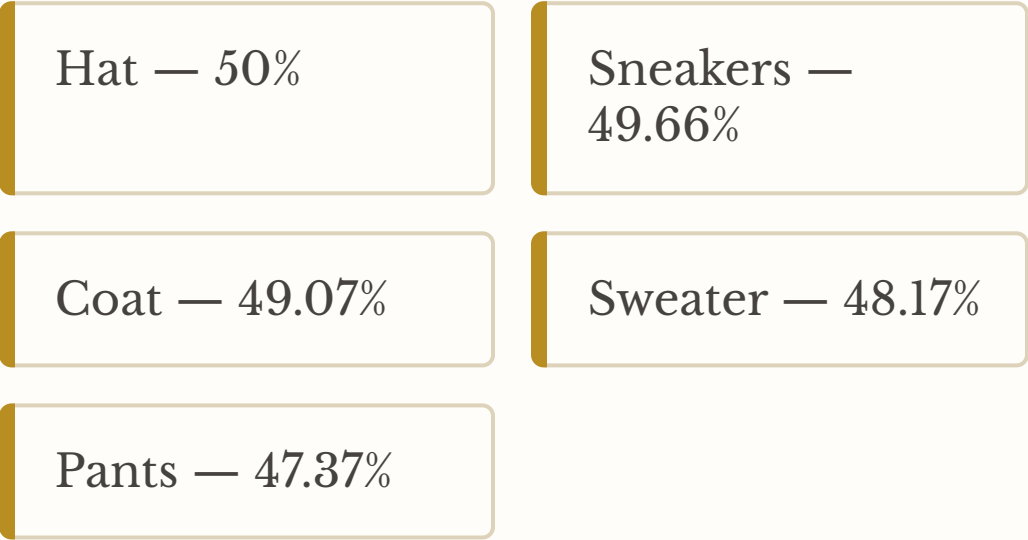
Male: \$157,890 · Female: \$75,191

Top Products & Discount Patterns

Top 5 Products by Rating

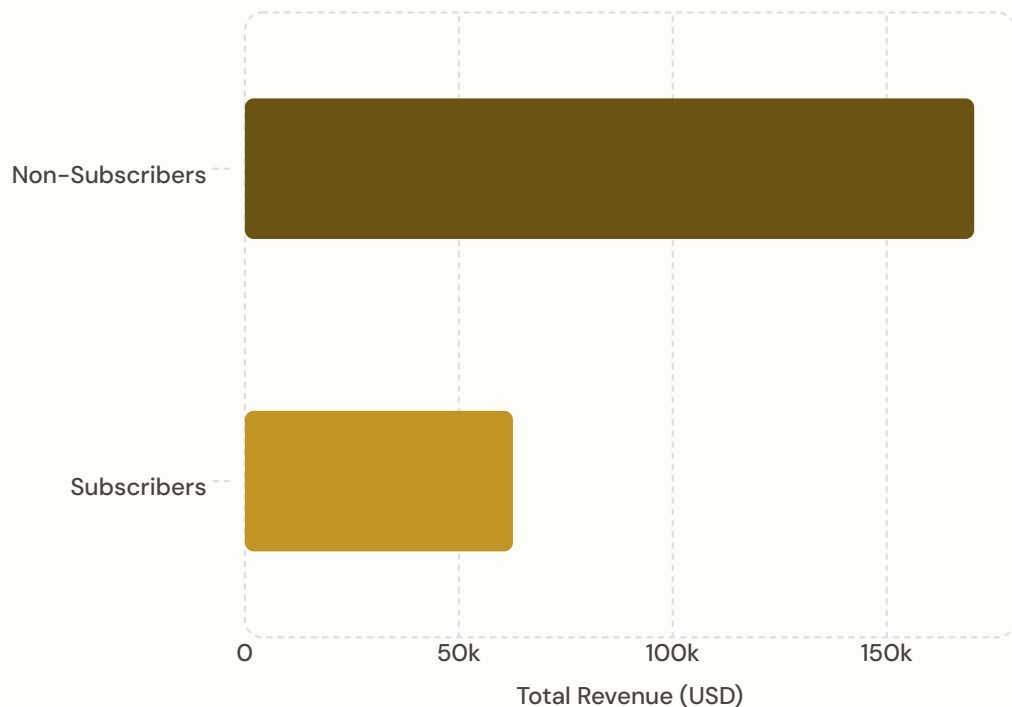


Most Discount-Dependent Products



Subscription Behavior

Subscription Status



Subscriber Profile

27%

Of customers are
subscribers

\$59.49

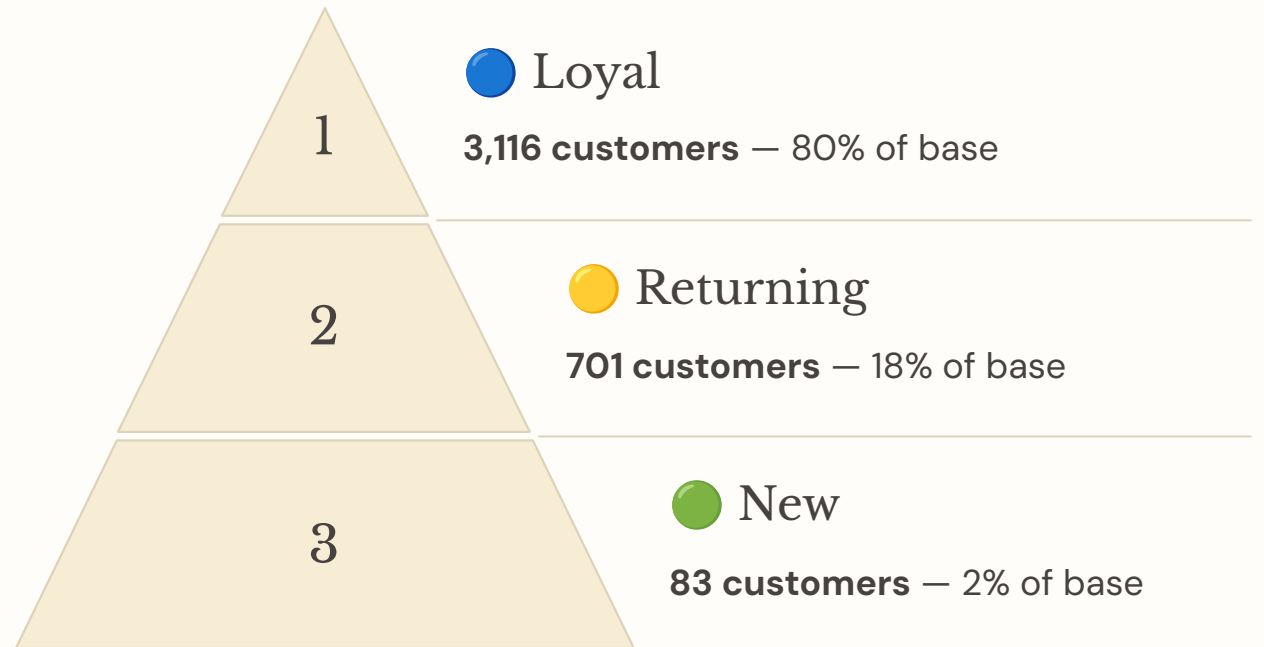
Avg spend (vs \$59.87
non-sub)

958

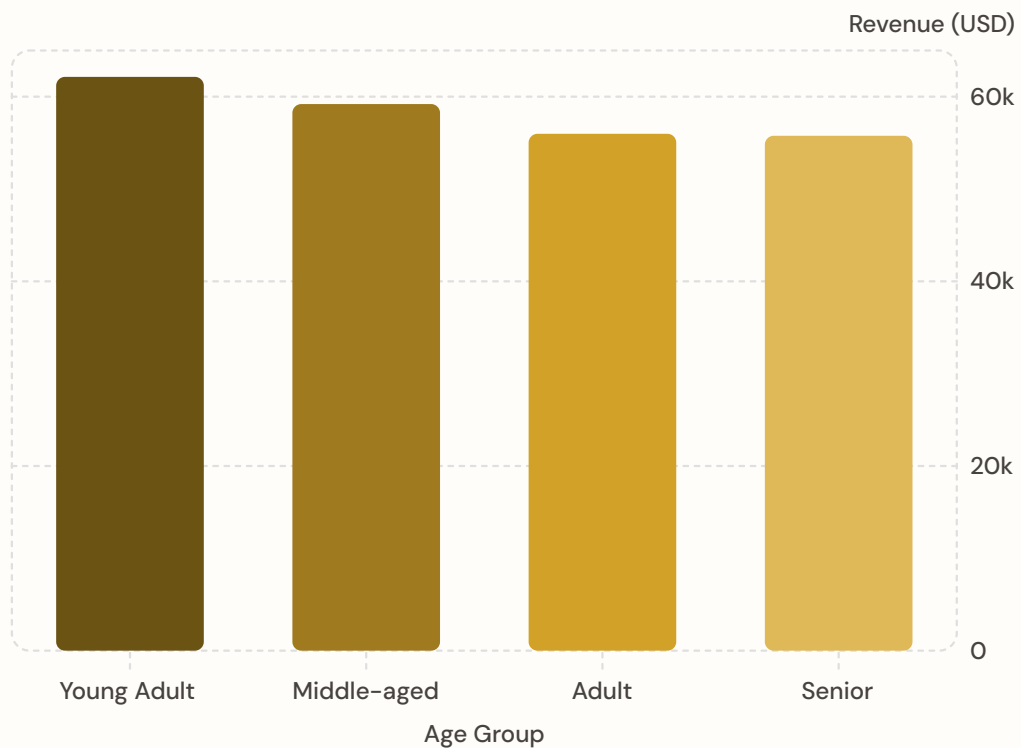
Repeat buyers who subscribe



Customer Segmentation



Revenue by Age Group



Insight

Young Adults lead in revenue contribution. Revenue declines modestly with age — all groups remain valuable targets.

Power BI Dashboard



Revenue by Category

Clothing leads at ~\$90K



Subscription Breakdown

27% subscribed vs 73% not



3.9K Customers

Avg purchase: **\$59.76**



3.75 Avg Rating

Across all products



Business Recommendations

1

Boost Subscriptions

Promote exclusive subscriber benefits

2

Loyalty Programs

Reward repeat buyers → Loyal segment

3

Review Discount Policy

Balance sales lift vs. margin

4

Product Positioning

Highlight top-rated products in campaigns

5

Targeted Marketing

Focus on high-revenue age groups & express shippers